

ECM3412

UNIVERSITY OF EXETER

**COLLEGE OF ENGINEERING, MATHEMATICS
AND PHYSICAL SCIENCES**

COMPUTER SCIENCE

Examination, May 2018

Nature-Inspired Computation

Module Leader: Dr Alberto Moraglio

Duration: TWO HOURS

Answer question 1, and two out of the other four questions.
Question 1 is worth 40 marks. Other questions are worth 30 marks each.

The marks for this module are calculated from 70% of the percentage mark for this paper plus 30% of the percentage mark for associated coursework.

No electronic calculators of any sort are to be used during the course of this examination.

This is a CLOSED BOOK examination.

SECTION A (Compulsory Question)**Question 1**

- (a) Write pseudocode for *particle swarm optimisation* (PSO). In your answer, you should include feasible parameter settings for each of the parameters associated with the algorithm.

(7 marks)

- (b) Most *evolutionary algorithms* use a mutation method to vary solutions during an optimisation run.

- (i) Explain why mutation operations must take into account the representation being used in the evolutionary algorithm.

(4 marks)

- (ii) Give an example of a situation where a mutation operator has been poorly chosen for a particular representation, and explain why the choice is poor.

(3 marks)

- (c) Craig Reynolds' *Boids* provide an efficient simulation of flocking behaviour on a computer. Name and briefly describe the three rules, known as *Reynolds' Rules*, used in this simulation.

(6 marks)

- (d) When using ant colony optimisation (ACO) to solve the travelling salesman problem (TSP), the transition rule $p_{ij}^k(t)$ for an ant k to go from city i to city j at the t -th iteration is calculated as:

$$p_{ij}^k(t) = \begin{cases} \frac{[\tau_{ij}(t)]^\alpha \cdot [\eta_{ij}(t)]^\beta}{\sum_{h \in J_i^k} [\tau_{ih}(t)]^\alpha \cdot [\eta_{ih}(t)]^\beta} & \text{if } j \text{ has not been visited} \\ 0 & \text{otherwise} \end{cases}$$

- (i) Explain each component of the above equation, i.e., J_i^k , $\tau_{ij}(t)$, $\eta_{ij}(t)$, α and β .

(4 marks)

- (ii) What are the effects of α and β ?

(2 marks)

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(e) In neural network training, one of the major reasons that lead to *overfitting* is an *overly complicated* neural network structure.

(i) Draw a diagram to illustrate the relationship between the *model complexity* and *error*, with respect to the *training error* and the *generalisation error*.

(4 marks)

(ii) Indicate in the diagram where the *under-fit region* and the *overfit region* are.

(4 marks)

(f) In Conway's "Game of Life", there are three major rules, i.e., *death*, *birth* and *stasis*. Explain in what circumstances a particular rule is triggered.

(6 marks)

(Total 40 marks)

SECTION B (Optional Questions)

Question 2

The term *evolutionary algorithms* includes a large number of algorithms that use the principle of evolution as their basis.

- (a) Write pseudocode for an *evolutionary algorithm*. In your answer, you should include feasible parameter settings for each of the parameters associated with the algorithm.

(6 marks)

- (b) Describe the main similarities and differences between *genetic algorithms* and *genetic programming*.

(6 marks)

- (c) Explain what is meant by the terms *direct* and *indirect* representations in evolutionary algorithms. Give an example of each.

(6 marks)

- (d) Evolutionary algorithms are sometimes criticised for not producing the same result every time they are run with a different random seed. Explain why evolutionary algorithms generate different solutions each time and why this is an important property on problems with high combinatorial complexity.

(6 marks)

- (e) The notion of convergence is an important one in evolutionary algorithms. Explain what is meant by convergence, and with the aid of a diagram, explain what is meant by premature/early convergence of the algorithm.

(6 marks)

(Total 30 marks)

Question 3

- (a) In multi-objective optimisation, what is a *Pareto set* and what is a *Pareto front*?

(2 marks)

- (b) It is well known that *convergence* and *diversity* are the cornerstones of multi-objective optimisation. To test the performance of a multi-objective optimiser, benchmark problems are designed with various difficulties concerning the convergence and diversity simultaneously.

- (i) Equation (1) gives a generic form of the ZDT benchmark function design:

$$\begin{aligned} &\text{minimize } \mathbf{F}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x})) \\ &\text{subject to } f_2(\mathbf{x}) = g(x_2, \dots, x_n)h(f_1(x_1), g(x_2, \dots, x_n)) \end{aligned} \quad (1)$$

where n is the number of variables. Explain which function controls the *convergence* and which one controls the *diversity*.

(2 marks)

- (ii) List three properties that make convergence difficult.

(3 marks)

- (iii) List three properties that make diversity difficult.

(3 marks)

- (c) Let us consider a multi-objective optimisation problem in which both objective functions are to be minimised. Figure 1, shown in the next page, gives a population distribution of 12 solutions in NSGA-II. By using the fast non-dominated sorting of NSGA-II,

- (i) how many non-dominated fronts can be obtained for this population?

(1 mark)

- (ii) Identify the solutions that belong to each non-dominated front.

(5 marks)

- (d) What is *many-objective optimisation*? Explain *four* reasons that make *many-objective optimisation* challenging.

(5 marks)

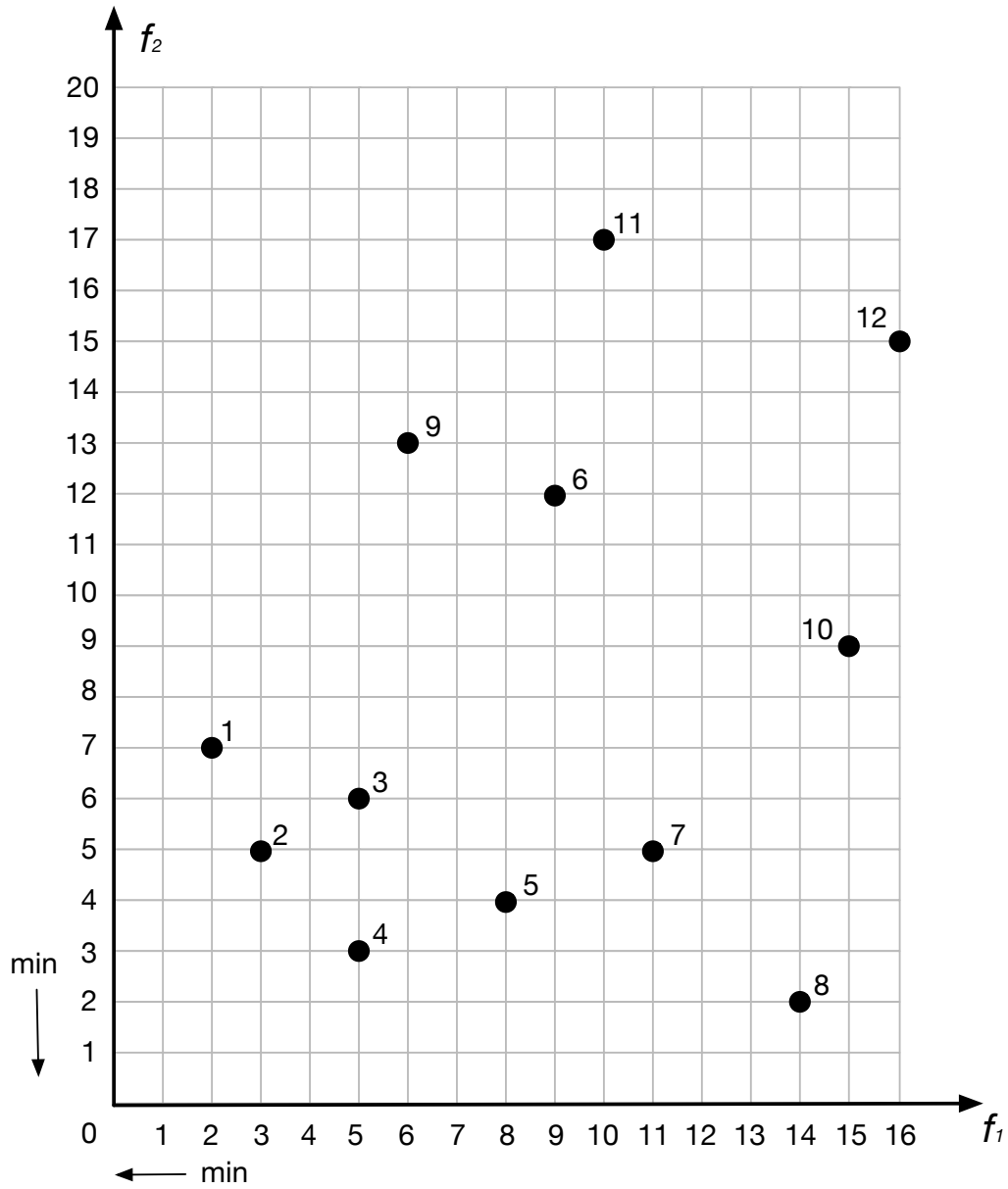


Figure 1: Population distribution in NSGA-II.

(e) Hypervolume (HV) is one of the most widely used performance metrics to evaluate the performance of an evolutionary multi-objective optimisation (EMO) algorithm. It is able to evaluate the convergence and diversity simultaneously.

- (i) Given a set of solutions $S = \{\mathbf{x}^1, \dots, \mathbf{x}^5\}$, and a reference point $\mathbf{z}^r$ as shown in Figure 2. In particular $\mathbf{x}^1 = (1, 9)^T$, $\mathbf{x}^2 = (3, 7)^T$, $\mathbf{x}^3 = (5, 4)^T$, $\mathbf{x}^4 = (7, 2)^T$, $\mathbf{x}^5 = (8, 1)^T$ and $\mathbf{z}^r = (9, 10)^T$, calculate the HV value of S , show the calculation process by using the diagram shown in Figure 2.

(6 marks)

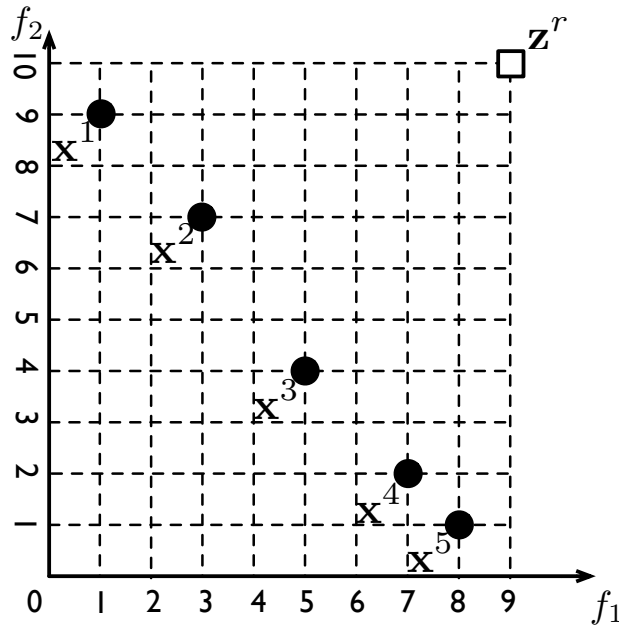


Figure 2: Population distribution

- (ii) Due to the theoretical property of HV, it has been widely used in the indicator-based EMO algorithm design. As for the population distribution shown in Figure 2, if we use HV to assign a fitness value to each solution like in SMS-EMOA, which solution should be removed at the end of an environmental selection? Why?

(3 marks)

(Total 30 marks)

Question 4

Neural networks are useful pattern recognition tools and can also be used to study human cognition.

- (a) Describe the neural network approach (including representation), and briefly the experimental findings in the paper Sejnowski, T. J. and Rosenberg, C. R. (1987) "Parallel networks that learn to pronounce English text" *Complex Systems* **1**.

(15 marks)

- (b) Table 1 gives the inputs $\mathbf{x}$ and output y values of an unknown mapping. Design a neural network to obtain such mapping. Draw a diagram to illustrate your design and validate it to get the correct mapping as shown in Table 1.

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	1

Table 1: The input and output table

(5 marks)

- (c) Figure 3 gives a simple example of three-layer neural network, where x_i represents an input neuron, y_i represents an output neuron and a_i represents a neuron in the hidden layer. In particular, w_{ij} denotes the weight associated with the connection between neurons in different layers. It is well known that, in the neural network training, the weight update rule is $w_{ji} := w_{ji} + \eta \delta_j x_i$ for the input layer while it is $w_{ji} := w_{ji} + \eta \delta_j a_i$ for the hidden layer, where η is the learning rate and δ_j is the back-propagation error at the j -th neuron.

- (i) Give the equation for the *error calculation* at an output layer node and a hidden layer node, respectively.

(6 marks)

- (ii) Accordingly, give the equation for calculating δ_4 and δ_8 .

(4 marks)

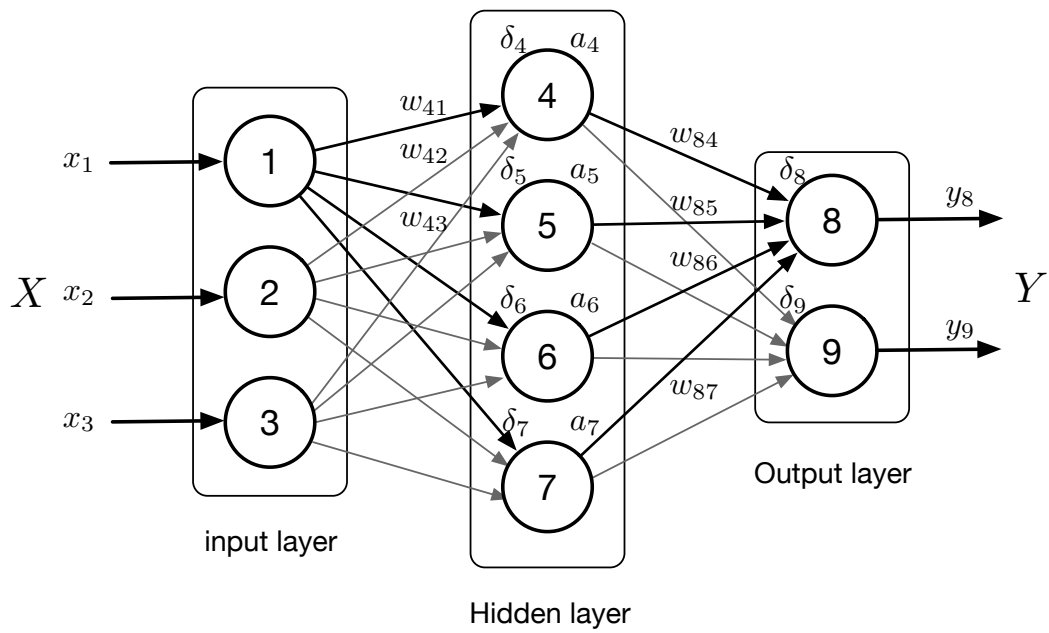


Figure 3: The back-propagation process of a neural network.

(Total 30 marks)